Pelvic Inflammatory Disease (PID)

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Abstract. Learn what PID is, its causes, diagnosis, complications, and how the microbiome may shape new therapies.

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Overview

Pelvic Inflammatory Disease (PID) is a significant infection of the female upper reproductive tract, affecting the uterus, fallopian tubes, ovaries, and other surrounding pelvic structures. It is most commonly caused by sexually transmitted infections (STIs), particularly *Chlamydia trachomatis* and *Neisseria gonorrhoeae*. These infections ascend from the cervix or vagina into the upper genital tract, where they trigger an immune-inflammatory response.[1] This inflammation can lead to severe long-term complications, such as infertility, ectopic pregnancy, and chronic pelvic pain.[2] PID is considered a major cause of morbidity among women of reproductive age, with millions of cases occurring globally each year. The condition is most prevalent in women aged 15 to 25 and is often linked to sexual behaviors, such as multiple sexual partners or inconsistent condom use.[3][4] According to the National Health and Nutrition Examination Survey (NHANES) 2013–2014, the prevalence of a self-reported lifetime diagnosis of PID was 4.4% among sexually experienced women of reproductive age, translating to approximately 2.5 million prevalent PID cases in women aged 18 to 44 years nationwide in the United States.[5] This prevalence varied significantly depending on sexual health history and sexual behaviors, with PID rates differing across different racial and ethnic groups. Recent research has highlighted the importance of the microbiome in the development and progression of PID.[6] The vaginal microbiota, when balanced, plays a protective role in preventing infections. However, disruptions, or dysbiosis, where harmful bacteria overgrow, can increase the risk of PID. This connection between the microbiome and PID highlights a potential area for novel diagnostic and therapeutic interventions. [7]

Associated Conditions

PID is not only a significant reproductive health issue but is also associated with other severe conditions, especially those affecting a woman's fertility. Infertility is one of the most concerning consequences of PID, as the inflammation and scarring of the fallopian tubes can prevent the passage of eggs from the ovaries to the uterus.[8] This condition can lead to tubal infertility, where the fallopian tubes are either blocked or impaired in their ability to function.[9][10] PID also increases the risk of ectopic pregnancies, which can be life-threatening if not treated promptly.[11][12]Chronic pelvic pain is a common long-term consequence, severely impacting a woman's quality of life.[13][14] Emerging research has also drawn links between PID and

endometriosis.[15] Studies have shown that women with a history of PID have a higher risk of developing ovarian cancer, especially in cases where the PID was chronic or recurrent.[16] The inflammatory environment caused by PID might contribute to carcinogenesis in the ovaries and fallopian tubes, as persistent inflammation can damage the epithelium of these organs, increasing the likelihood of malignant transformations.

Causes

PID is primarily caused by infections that ascend from the cervix and vagina into the upper genital tract. The most common pathogens implicated in PID are *Chlamydia trachomatis* and *Neisseria gonorrhoeae*, both of which are sexually transmitted infections (STIs).[17] These bacteria can trigger an immune response in the reproductive organs, leading to inflammation and, if untreated, long-term damage to the fallopian tubes, ovaries, and uterus. Other bacteria, including those associated with bacterial vaginosis (*Peptostreptococcus* and *Bacteroides* species) and enteric pathogens such as *Escherichia coli*, can also contribute to PID, especially in post-menopausal women or those with other underlying health conditions.[18][20] The pathogenesis of PID involves an immune-inflammatory response where the body attempts to fight off the infection, but in the process, it inadvertently damages the reproductive tissues, leading to scarring and adhesions that disrupt normal reproductive function.

Diagnosis

Diagnosing PID is primarily clinical, as it relies heavily on patient history and physical examination. Women with PID often present with lower abdominal or pelvic pain, vaginal discharge, dyspareunia (pain during intercourse), and symptoms of genital tract inflammation.[21] However, the variability in symptoms makes PID difficult to diagnose accurately, especially in cases where the infection is subclinical or presents with mild symptoms. Diagnosis is confirmed through clinical findings such as cervical motion tenderness, uterine tenderness, and adnexal tenderness, often accompanied by abnormal vaginal discharge. Laboratory tests, including nucleic acid amplification tests (NAAT) for *Chlamydia trachomatis* and *Neisseria gonorrhoeae*, are essential for confirming the infectious causes of PID.[22] In cases where the diagnosis is uncertain, or when complications such as tubo-ovarian abscess are suspected, imaging studies such as pelvic ultrasound may be used, or when complications are suspected, laparoscopy is considered a gold standard for confirming PID.[23][24] Advancements in diagnostic techniques are also exploring the potential of microbiome signatures and metabolomic profiles to detect PID, offering a non-invasive and more precise way to diagnose and monitor the disease.[25]

Primer

PID is a disease involving the female reproductive organs, immune system, and microbiome. Understanding the microbiome's role in PID is important, as the gut and vaginal microbiota are deeply interconnected with immune functions and inflammation.[26] The immune-inflammatory response triggered by infections in the genital tract can have wide-reaching effects, potentially leading to long-term complications like infertility, endometriosis chronic pelvic pain, and an increased risk of ovarian cancer.[27] Studies increasingly focus on how manipulating the microbiome could provide novel treatment avenues for PID, including probiotics, fecal microbiota transplantation, vaginal microbiota transplantation, and other microbiome-targeted interventions. Understanding PID as a disease that goes beyond simple infection to

involve immune system dysfunction and microbiome imbalance is vital for developing effective diagnostic and therapeutic strategies.

Metallomic Signature

Trace metals and minerals play a significant role in the pathophysiology and management of pelvic inflammatory disease. NHANES reveal a notable association between dietary intake of certain trace metals, for example, copper and magnesium, and PID risk. Low dietary copper intake correlates strongly with increased PID prevalence, suggesting that adequate copper levels may contribute to regulating inflammatory responses associated with PID.[28] Similarly, magnesium intake is inversely related to PID risk, potentially through its involvement in cellular processes like DNA synthesis, protein production, and modulation of systemic inflammation.[29] These minerals influence immune function and oxidative stress, which are critical in PID's chronic inflammation and tissue damage.

Metabolomic Signatures

In PID, inflammatory responses triggered by ascending infections lead to oxidative stress and immune dysregulation, which are reflected in changes in metabolites such as amino acids, lipids, and bioactive small molecules. Key metabolites involved include arginine and ornithine, central to the arginine-proline metabolic pathway, that regulate nitric oxide synthesis, a critical mediator of inflammation and tissue injury in PID.[30] Elevated arginine levels correspond with increased nitric oxide synthase (NOS) activity, exacerbating oxidative damage and contributing to chronic inflammation.Lipid metabolites, including leukotrienes and arachidonic acid derivatives, also play significant roles in modulating inflammatory signaling, promoting vasodilation, leukocyte recruitment, and cytokine release within the pelvic tissues.[31] Disruptions in glutathione metabolism, a primary antioxidant defense, indicate heightened oxidative stress and compromised cellular detoxification during PID.

Interventions

We assess microbiome-targeted interventions (MBTIs) by analyzing their effect on the gut, vaginal and pelvic microbiota, clinical symptom resolution, and the modulation of bacterial pathogens responsible for PID. This method classifies interventions as validated, promising, or experimental based on the strength of scientific evidence linking changes in microbial composition to improved clinical outcomes, such as reduced inflammation, infection, and recurrence of PID. Focusing on restoring microbial balance, enhancing mucosal immunity, and inhibiting pathogenic overgrowth helps identify interventions that are most likely to improve patient outcomes and prevent long-term complications associated with PID.

Intervention	Classification	Mechanism of Action	MBTI Status
Probiotics	Supplement	Lactobacillus species restore vaginal microbiota, inhibiting pathogen colonization, modulating vaginal pH, and enhancing mucosal immunity, thus reducing inflammation and susceptibility to PID-associated pathogens such as *Chlamydia trachomatis* and *Neisseria gonorrhoeae*.[32][33]	Validated
Lactoferrin	Supplement	Lactoferrin exhibits prebiotic and antimicrobial activities, stimulating growth of beneficial vaginal microbiota while suppressing pathogens.[34] It modulates inflammation and strengthens mucosal immunity, effectively reducing PID severity and complications linked to microbial dysbiosis.[35]	Promising Candidate
Vaginal Microbiota Transplantation (VMT)	Microbiota-based therapy	The transfer of healthy vaginal microbiota restores beneficial bacterial dominance, thereby re-establishing microbial homeostasis, preventing pathogen colonization, and mitigating inflammatory responses characteristic of PID.[36]	Experimental
Fecal Microbiota Transplantation (FMT)	Microbiota-based therapy	Similar to VMT, the transfer of beneficial gut microbiota restores gut-vaginal microbial axis integrity, thereby reducing systemic inflammation, enhancing mucosal immunity, and indirectly ameliorating PID symptoms through modulation of intestinal flora and host immune responses.[37]	Experimental
Metformin	Parmaceutical	Metformin modulates gut microbiota, improves mucosal barrier function, and reduces systemic inflammation via activation of AMP-activated protein kinase (AMPK), potentially decreasing inflammatory cytokines and oxidative stress markers involved in PID pathogenesis.[38][39]	Promising Candidate
Bacteriophage Therapy	Biological Therapy	Bacteriophages provide a precise antibacterial effect without disrupting beneficial microbiota. This targeted approach prevents bacterial resistance, reduces inflammation, and restores microbial balance within the reproductive tract.[40]	Experimental
Antimicrobial Peptides	Biological Therapy	The use of naturally derived or synthetic peptides that possess antimicrobial properties, targeting pathogenic bacteria associated with PID.[41] These peptides modulate vaginal	Promising Candidate

		microbiota composition, support mucosal immunity, reduce microbial colonization linked to PID pathogenesis.[42]	
Hyperbaric Oxygen Therapy (HBOT)	Physical Therapy	HBOT enhances oxygen delivery and reduces inflammation, promoting beneficial microbial growth and suppressing pathogenic anaerobic bacteria linked to PID.[43]	Under Investigation

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